

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.





Dataset at a Glance

3,900

Purchases

Across all categories

18

Columns

Demographics, behavior, purchases

37

Missing Values

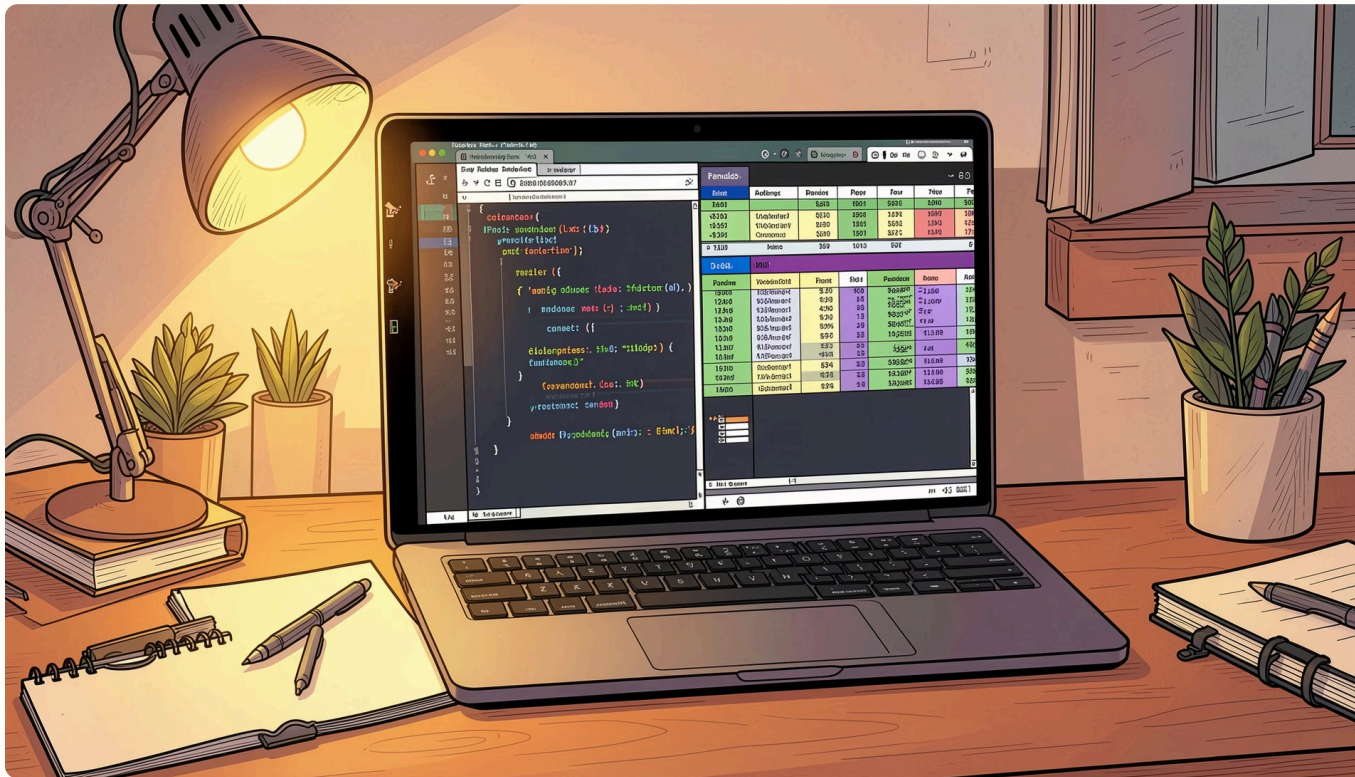
In Review Rating only

50

Locations

Geographic coverage

Data Preparation in Python



→ Clean & Standardize

Imputed missing ratings by category median; renamed to snake_case

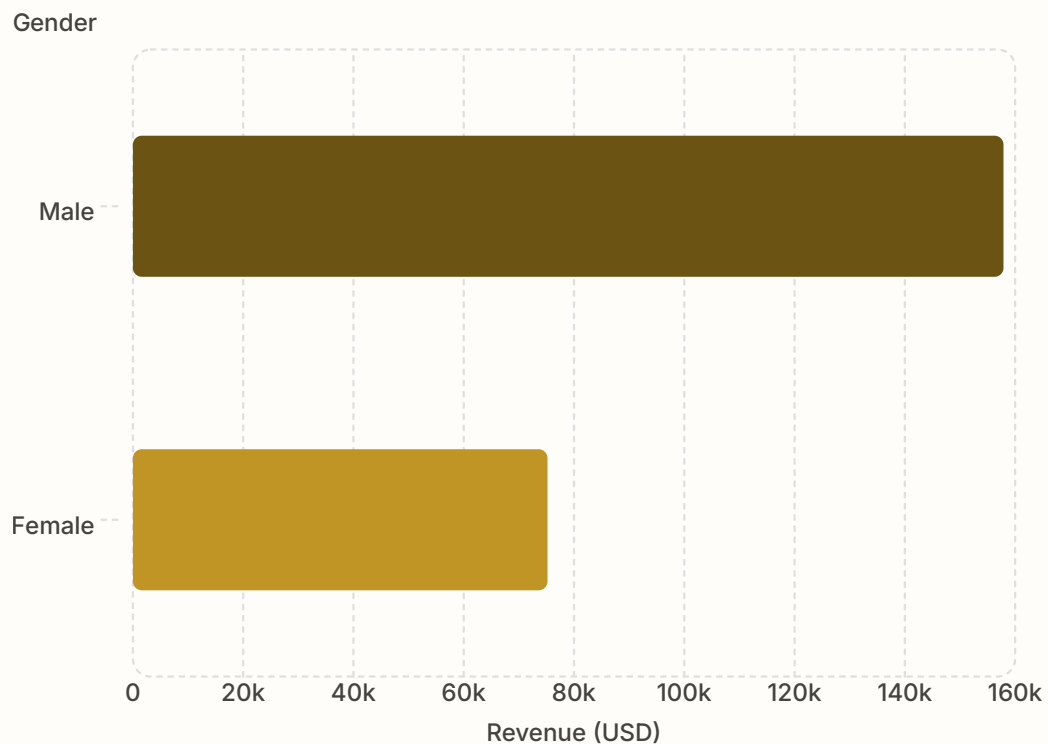
→ Feature Engineering

Created `age_group` and `purchase_frequency_days`

→ Load to PostgreSQL

Cleaned DataFrame loaded for SQL analysis

Revenue by Gender



Key Finding

Male customers generated **2.1× more revenue** than female customers — **\$157,890 vs \$75,191**.

Products, Discounts & Shipping

Top Rated Products

Gloves (3.86★), Sandals (3.84★),
Boots (3.82★)

Most Discounted

Hat (50%), Sneakers (49.7%), Coat
(49.1%)

Shipping Spend

Express: \$60.48 avg vs Standard:
\$58.46



Subscribers vs. Non-Subscribers



73% Non-Subscribers

2,847 customers · \$170K revenue

27% Subscribers

1,053 customers · \$62.6K revenue

Average spend is nearly identical — **subscription is an untapped growth lever.**

Customer Segmentation

3,116

Loyal

80% of all customers

701

Returning

18% of all customers

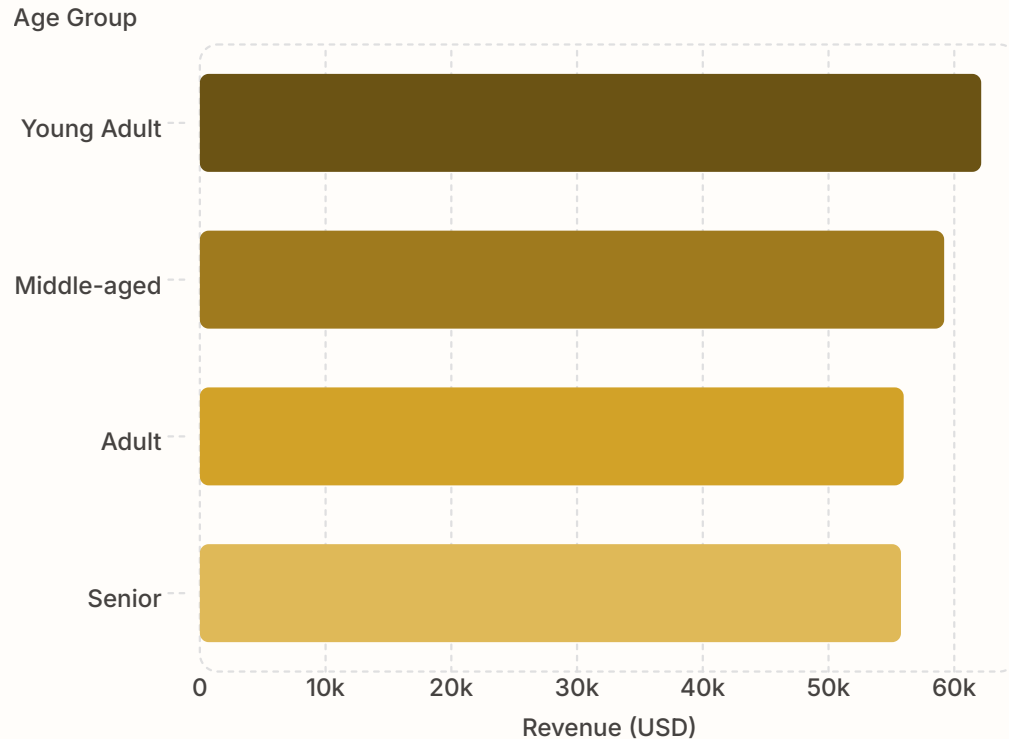
83

New

2% of all customers

Of customers with **>5 purchases**, 958 are subscribers vs. 2,518 non-subscribers — loyal buyers are **less likely to subscribe**.

Revenue by Age Group & Category



Top Products by Category

- **Clothing:** Blouse, Pants, Shirt
- **Footwear:** Sandals, Shoes, Sneakers
- **Outerwear:** Jacket, Coat
- **Accessories:** Jewelry, Sunglasses, Belt

Young Adults lead revenue — **\$62,143**. Clothing dominates all categories.

Power BI Dashboard

Key Metrics

3.9K customers · \$59.76 avg
purchase · 3.75★ avg rating

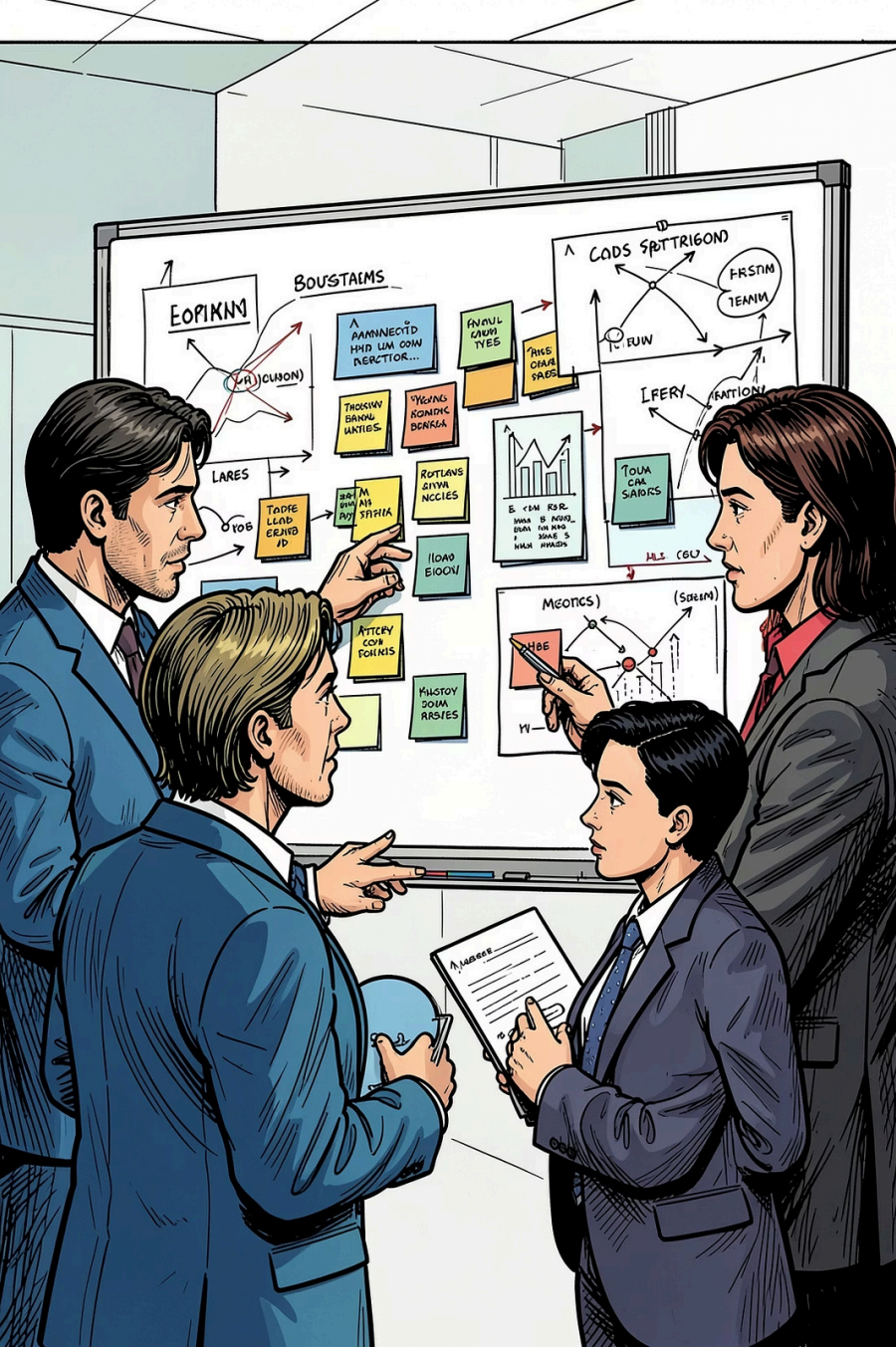
Filters

Subscription, Gender, Category,
Shipping Type

Visuals

Revenue & Sales by Category and Age Group





Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow the Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups and express-shipping users